

IEEE 1687 / 1838 Backup Evidence

12

BACKUP PAGES

EVIDENCE GUIDE

Detailed checkpoints, evidence, and scope boundaries for Steps 1–12.



Verified: Steps 1–7 | Next: Steps 8–9 local ATPG | Bonus: Steps 10–11 | Step 12: cleanup

Step 1 — Capture Working Run Evidence



STAGE 1

Run Environment

ACTION

Run current flow

- Record RTL, libraries, file list, and Tessent version
- Execute the documented Tcl sequence from the working folder
- Record the design-specific edits used for this case

PASS CRITERIA

Must pass

- ✓ RTL elaborates cleanly
- ✓ Required Tcl sequence completes without fatal error
- ✓ ICL, patterns, logs, and reports are generated

CAPTURE

Keep

Command + final log

version · run order · PASS line

Folder tree

Tcl files + case-specific edits

Outputs

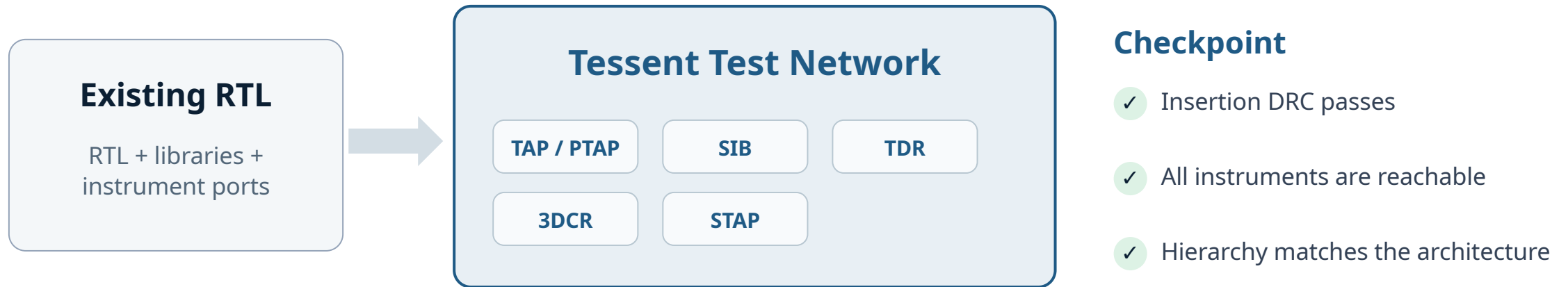
RTL · ICL · patterns · reports

Step 2 — Prove Insertion and Reachability



STAGE 2

RTL Insertion



Evidence

Insertion DRC report

Pre/post hierarchy or schematic

Connectivity / reachability report

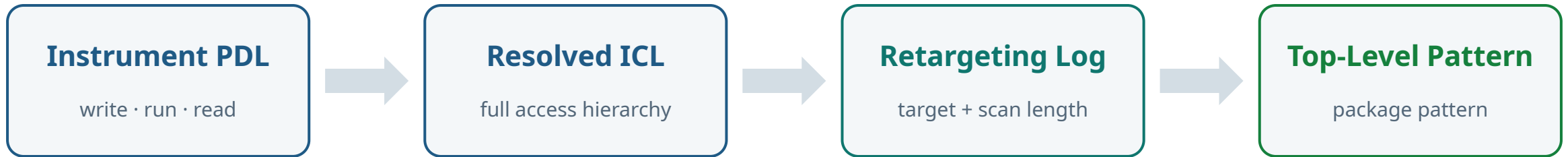
Insertion is complete only when reachability is proven.

Step 3 — Prove Full-Path PDL Retargeting



STAGE 3

PDL Retargeting



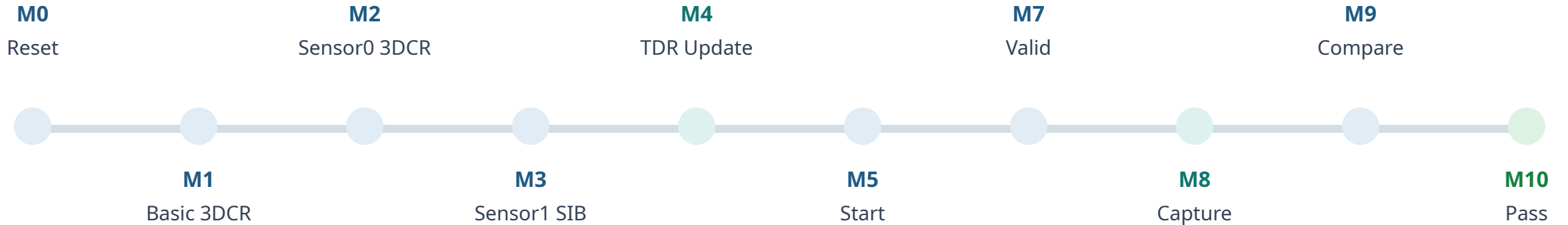
Checkpoint

- ✓ ICL loads without errors
- ✓ Target path is fully resolved
- ✓ Scan length and pattern count are recorded

Evidence to capture

PDL operation · resolved ICL path · retarget log · package-pattern excerpt

Step 4A — Read the RTL Waveform by Event



Markers isolate access, selection, or response failures. Confirm M6 from the testbench before labeling it.

Step 4B — Prove All Three Access Scopes



01

IEEE 1687

TAP → SIB → TDR → Instrument

Local PASS · mismatch = 0



02

1687 + IEEE 1838

PTAP → 3DCR / STAP → SIB → TDR

1838 wrapper PASS ·
mismatch = 0



03

stack_top

Package → Basic STAP → Sensor STAP → SIB → TDR

Stack forward / return PASS ·
mismatch = 0



Capture one expected/observed result summary for each access scope.

Step 5 — Prove the Test Network Survives Synthesis



STAGE 5

Synthesis



Checkpoint

- ✓ Test network survives optimization
- ✓ Test clocks, resets, and constraints are applied

Evidence

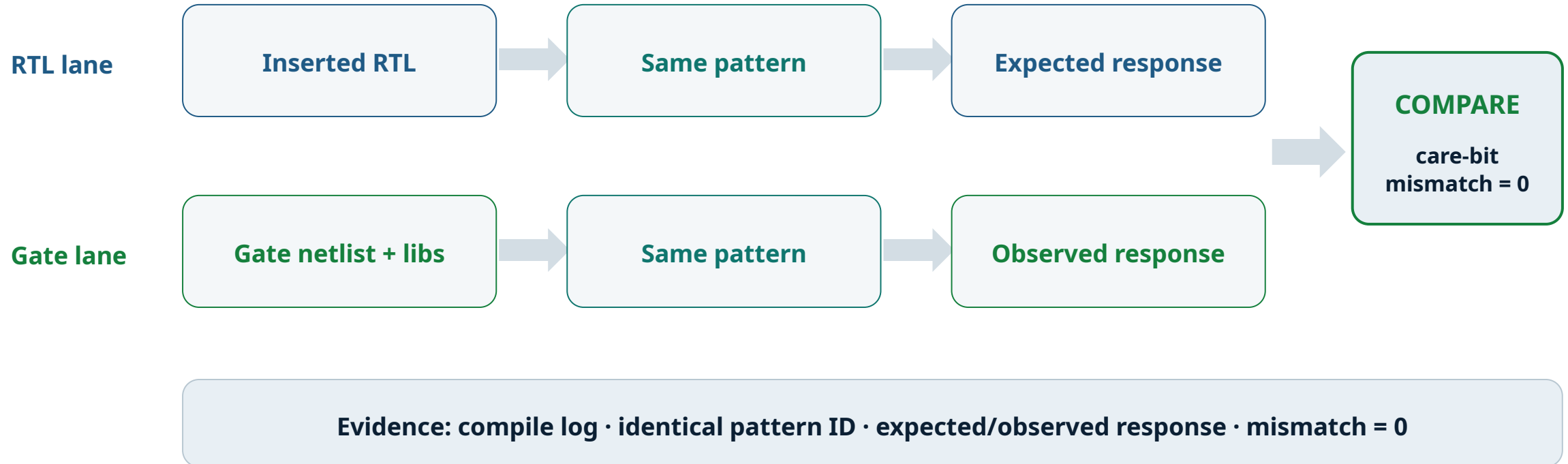
- Pre/post hierarchy or instance comparison
- Synthesis log · dont_touch / SDC · gate netlist

Step 6 — Compare RTL and Gate with the Same Pattern



STAGE 6

Gate Verification

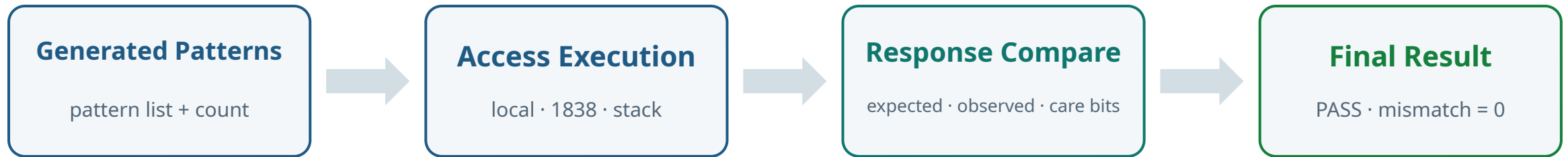


Step 7 — Auto-Generated Pattern Verification Is Complete



EVIDENCE
CHAIN

Generated Pattern Verification



Checkpoint

- ✓ Generation log and pattern count are recorded
- ✓ The same set completes at selected access scopes
- ✓ Final compare reports care-bit mismatch = 0

Evidence to keep

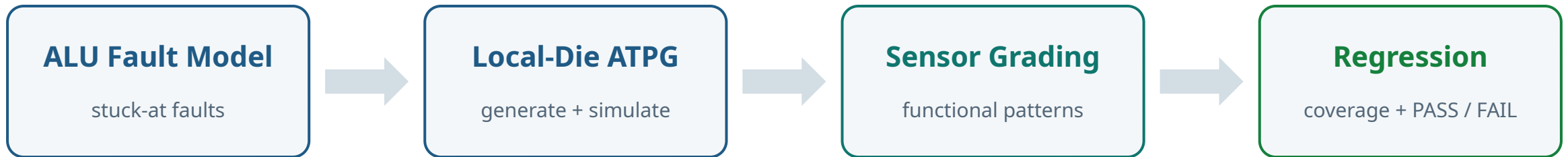
**Generator log · pattern manifest ·
simulation log · final PASS summary**

Steps 8-9 — Keep ATPG Coverage at Local-Die Scope



NEXT
MILESTONE

Local-Die ATPG, Fault Grading & Regression



Checkpoint

- ALU ATPG coverage is reported
- Sensor functional patterns are fault graded
- Regression reports coverage, aborted faults, and mismatch

Scope boundary

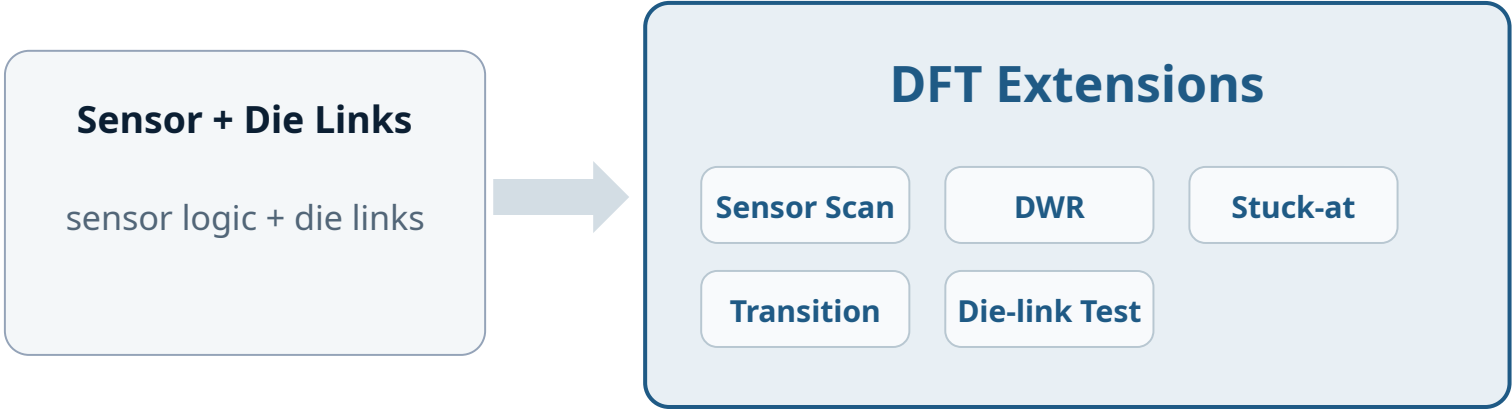
Run one representative ATPG pattern through `stack_top` only as an end-to-end sanity check.

Steps 10-11 — Optional Sensor and Die-Link Extensions



IF
EXECUTED

Sensor Scan & DWR Interconnect Test



Checkpoint

- Sensor structural coverage reported
- DWR tests die-to-die faults
- Run only if schedule and die-link ports permit

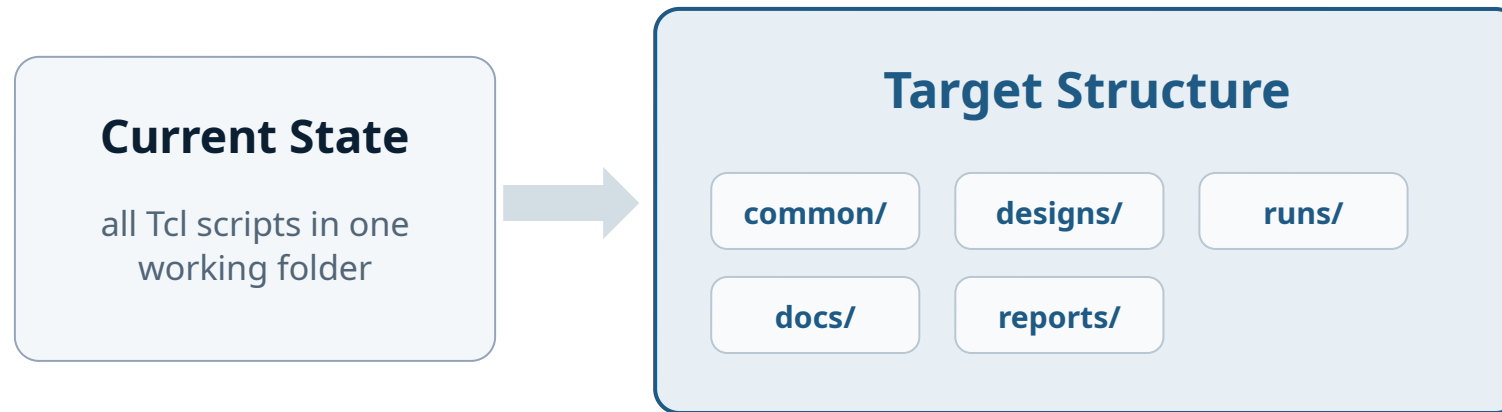
Evidence if executed	Scan + ATPG report	DWR coverage	Die-link simulation
	Additional bonus does not block the main local-die ATPG deliverable.		

Step 12 — Organize the Working Tcl Environment

1 INVENTORY — 2 COMMON FLOW — 3 DESIGN CONFIG — 4 RUN FOLDERS — 5 CHECKS — 6 HANDOFF

STEP 12 ·
PLANNED

Environment Consolidation



Checkpoint

- Common flow is separated from design-specific settings
- One documented entry point and run order are defined
- A clean output folder reproduces the same key results

Evidence

Folder tree

Config + entry script

Clean-run log + checklist

Clean rerun = new output folder, no manual copying or edits, same DRC / pattern / mismatch results.